Experimental Comparison of Inclined Flows with and without Intense Sediment Transport: Flow Resistance and Surface Elevation

Daniel Rebai[1]; Diego Berzi[2]; Francesco Ballio[3]; and Vaclav Matousek[4]

Abstract: The effect of the intense transport of lightweight sediment without turbulent suspension (Shields numbers of mixture flow from 0.34 to 1.80) on resistance and surface elevation of water flow in an open channel is experimentally examined. Flows over fixed and mobile plane beds, at the same water flowrate and bed slope, are compared. Detailed measurements—including distributions of solids and velocity—permit to identify flow interfaces and determine the role of sediment transport on flow resistance. Two friction factors for mobile bed flows are introduced: one for water only and another one for the mixture. The first seems to be insensitive to the water submergence, while the latter follows the classic formula for the hydraulically rough regime: both increase with respect to the equivalent clear water flow. The water surface elevation seems unaltered by the presence of sediment transport, thus reducing the need for the modelling of the evolution of an erodible bed surface in, at least, some routine applications in fluvial hydraulics. **DOI: 10.1061/(ASCE)HY.1943-7900.0002024.** *© 2022 American Society of Civil Engineers.*

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Introduction

From an engineering perspective, the problem of sediment transport in open channels has been historically reduced to determine: (1) a relationship between the water flow depth, water flow rate, and bed slope (friction law); and (2) a relationship between the solid flow rate and a measure of the intensity of the bed shearing by water flow (sediment transport formula) (Du Bois 1879; Donat 1929; Schoklitsch 1934; Einstein 1950; Ferguson 2007; Ferguson 2012; Recking et al. 2008a). Both relationships are usually derived in steady and fully developed flow configurations, but can also be applied to inhomogeneous and unsteady situations and are widely adopted in commercially and freely available numerical codes. It soon became clear (Meyer-Peter and Müller 1948; Smart and Jaeggi 1983) that the resistance encountered by the flowing water is affected by the amount of moving sediments, which in turn depends on the intensity of shearing (Bergeron and Carbonneau 1999; Campbell et al. 2005; Recking et al. 2008b; Simoes 2010).

The concept of flow resistance is widely discussed in the literature (Yen 2002), where it is defined as either "*the mechanical drag (or friction) exerted on the flow by the rough channel perimeter*" (mechanical description) or as "*the potential and kinetic energy expended by the river in overcoming the friction of the bed and banks*" (thermodynamic description). When made dimensionless, both descriptions are indeed equivalent [see the later Eq. (1)] and proportional to the friction factor, which is the dimensionless shear stress exerted by the bed on the flow (Song et al. 1998; Abrahams and Li 1998; Hu and Abrahams 2004, 2005; Habibzadeh and Omid 2009).

A fair question would be whether the presence of sediment transport increases or decreases the friction factor with respect to an equivalent flow with no particle motion. In the Sedimentation Manual, Marcelo Garcia summarized the general consensus of the sediment transport community on this topic as follows: "*at high bed shear stresses and sediment transport intensities in sand-bed streams, dunes are washed out and the bed becomes plane (. . .) flow resistance increases drastically with flow intensity in this regime*" (Garcia 2008). However, previous works have contradicted each other, suggesting that sediment transport either increases (Abrahams and Li 1998; Banerjee et al. 2019; Ferro 2018; Gao and Abrahams 2004; Hou et al. 2019; Hu and Abrahams 2004, 2005; Recking 2006; Recking et al. 2008a; Rickenmann and Recking 2011; Song et al. 1998), decreases (Hanes 1986; Carbonneau and Bergeron 2000; Nikora and Goring 2000; Dey et al. 2012), or leaves unaltered (Campbell et al. 2005) the friction factor.

There are a number of reasons why there is no general agreement on the role of sediment transport on the flow resistance. First of all, there is a degree of freedom while defining equivalence of pairs of flows with and without sediment transport, as one can fix only two of the three quantities that determine the friction law (depth, flow rate, or bed slope).

Second, as mentioned, the friction factor is usually defined as the dimensionless shear stress exerted by the bed. This definition implies many nontrivial issues to be addressed: which shear stress should be considered; how it should be made dimensionless; how it should be measured; and where (and what) is the bed. Indeed, the first two issues are part of a more fundamental question about the physical nature of sediment transport. That is, whether this phenomenon should be treated in the context of a single-phase fluid or a two-phase mixture.

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For weak sediment transport (Meyer-Peter and Müller 1948), there are only a few moving particles so that the bed shear stress and the density and cross-sectionally averaged velocity used to make it dimensionless in turbulent flows are unambiguously those of the fluid (water in our case). For more intense sediment transport in which the number of moving particles is significant (Jenkins and Hanes 1998), one should, in principle, distinguish between the fluid and the particle densities, stresses, and velocities; hence, the dimensionless shear stress can involve quantities pertaining to the fluid, particles, or mixture. To make everything murkier, many works in the literature have adopted ambiguous approaches, mostly for experimental convenience, such as estimating the cross-sectionally averaged velocity using the water flow rate and the mixture cross-section and/or making the mixture bed shear stress dimensionless using the water density (Abrahams and Li 1998; Banerjee et al. 2019; Bergeron and Carbonneau 1999; Best et al. 1997; Calomino et al. 2004; Campbell et al. 2005; Dey and Raikar 2007; Dey et al. 2011, 2012; Hanes 1984, 1986; Hou et al. 2019; Hu and Abrahams 2004; Nikora and Goring 2000; Recking et al. 2008a; Rickenmann 1991; Simoes 2010; Smart and Jaeggi 1983; Song et al. 1998; Williams 1970). The experimental measurement of the bed shear stress is also not straightforward, even for steady and fully developed situations. It can be estimated from mean velocity (Clauser 1954) and Reynolds shear stresses (Revil-Baudard et al. 2015) profiles; or it can be deduced from the slope of energy grade line and the flow depth (Dey 2014). However, the three methods are not always consistent with each other (Dey et al. 2012), with the additional problem that, more often than not, it is unclear whether the measurements refer to the water, sediments, or both.

Finally, the location of the bed itself, that is the position of the interface between what is considered flowing and what is immobile is still an unsolved problem. Obviously, if sediment transport takes place over a fixed, rigid base where the particles experience a significant slip (Abrahams and Li 1998; Bergeron and Carbonneau 1999; Best et al. 1997; Campbell et al. 2005; Carbonneau and Bergeron 2000; Hou et al. 2019; Hu and Abrahams 2004), the identification of the bed is trivial. Not quite so in the case of sediment transport over an erodible bed, that is a loose assembly of particles that can be incorporated into the flow (Banerjee et al. 2019; Dey and Raikar 2007; Hu and Abrahams 2004; Nikora and Goring 2000; Recking 2006; Recking et al. 2008b; Song et al. 1998). No macroscopic discontinuities manifest at the interface between the flow and the erodible bed: as such, its determination is often arbitrary.

From an experimental point of view, the most adopted approach is searching for the region where the measured velocity vanishes, but it has been shown (Komatsu et al. 2001) that the velocity in the bed approaches zero only asymptotically. A more physically-sound criterion (Berzi et al. 2019) characterizes the bed as the region in which the particles develop a network of persistent contacts and associated rate-independent components of the stresses, with the local particle volume concentration (particle volume over total volume) exceeding a critical value. However, experimentally measuring the concentration is challenging, even more so in the denser part of the flow (Capart and Fraccarollo 2011), and the critical value is known only in the case of spherical particles (Chialvo et al. 2012).

The research question of the present work is twofold. From a conceptual point of view, we investigate the effect of sediment transport on the friction factor; for this purpose, special care is devoted to the definition of unambiguous and appropriated quantities describing the system. Then, we discuss the modelling implications of our findings. By doing so, we focus on steady and fully developed, free surface flows of water in a rectangular, inclined channel either over a rough, rigid base in the absence of sediment transport or over erodible beds in the presence of intense sediment transport, without turbulent suspension. We choose the water flow rate and the slope of the bed as control variables and define pairs of equivalent conditions by setting equal values for such parameters. Effects of sediment transport are evaluated on the resulting water depths and friction factors. Water flow rate and bed slope are the two quantities that are easier to control in the experiments. Also, unlike the water depth, they are not affected by the uncertainty associated with the location of the erodible bed.

Conceptual Framework

We focus on steady and fully developed, turbulent, free surface flows of water of density ρ_w and viscosity μ in a rectangular channel of width W and angle of inclination ϕ with respect to the horizontal direction, with g as the gravitational acceleration. The water can flow either over an erodible bed in the presence of sediment transport or over a rigid, rough bed in the absence of sediment transport (Fig. 1). In both cases, we take x and y to be the direction parallel and perpendicular to the bed, respectively, and neglect variations in the span-wise direction. As mentioned in the Introduction, the two flow configurations of Fig. 1 are equivalent, in the sense that the angle of inclination of the bed and the water flow rate per unit channel width, q_w, are the same. This also implies that the stream power per unit basal area, $\rho_w g q_w \sin\phi$, associated with the water is the same in the two configurations.

For flows over erodible beds, sediments are characterized by an average equivalent diameter d (the diameter of a sphere having the same volume) and density ρ_s. The erodible bed and the free surface are located at $y = 0$ and $y = H_M$, respectively [Fig. 1(a)].

For flows over rigid, rough beds, the roughness originates from a layer of the same particles of the erodible bed scenario glued at a flat plate. The rigid bed and the free surface are located at $y = \xi$ and at $y = H_F$, respectively [Fig. 1(b)].

The offset ξ is determined by imagining that the volume of sediments per unit basal area flowing over the erodible bed in the configuration of Fig. 1(a), i.e., $\int_0^{H_M} \nu dy$, where ν is the local particle volume concentration, is piled up at rest under the rigid base. Hence, $\nu_r \xi = \int_0^{H_M} \nu dy$, where ν_r is the concentration at rest. This is like assuming that the flow of Fig. 1(b) takes place over an

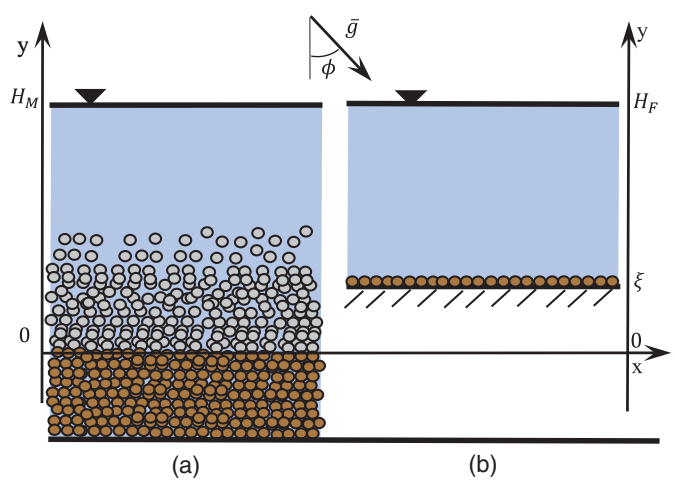

Fig. 1. (Color) Sketches of the flow configurations with the frame of reference of (a) flow over an erodible bed in the presence of sediment transport; and (b) flow over a rigid, rough bed in the absence of sediment transport.

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impermeable granular substrate, whose volume is equal to the volume of flowing sediments of Fig. 1(a). The two flow configurations are now equivalent also in the sense that the volume of sediments is the same. This strategy ensures that the location of the free surface in both scenarios is representative of the actual water surface elevation with respect to a common reference level, valuable information from an engineering perspective. We should keep in mind, however, that if the flow of Fig. 1(b) would take place over a fixed *permeable* granular substrate, the resulting water surface elevation, at the same values of q_w and ϕ, would be lower, because part of the water volume would be allowed to flow in the substrate itself (Manes et al. 2011).

From a thermodynamic point of view, the flow resistance is given by the stream power subtracted per unit basal area, i.e., the rate at which the mechanical energy is taken from the flow. From a mechanical point of view, the flow resistance is given by the forces per unit basal area that act against the motion. In steady and fully developed, gravity-driven, free surface flows, the energy taken from the flow is the potential energy associated with the gravitational field, and the forces that resist the flow must equal the component of the weight in the direction parallel to the flow itself. Then, in dimensionless terms, we obtain

$$\frac{\rho g q \sin\phi}{\rho U^3} = \frac{\rho g H \sin\phi}{\rho U^2} = \frac{f}{8} \tag{1}$$

where the first term represents the dimensionless stream power per unit basal area and the second term is the dimensionless weight component in the flow direction. Here, ρ, q, H, and U are some characteristic density, flow rate per unit channel width, depth, and cross-sectionally averaged velocity, respectively. As shown, when the flow resistance is made dimensionless using quantities appropriated to inertial flows (density and cross-sectionally averaged velocity), the thermodynamic and mechanical points of view coincide, and are embedded in the friction factor f.

In the case of flows over rigid, rough beds in the absence of sediment transport, the physical meaning of and the choice of the appropriate quantities in Eq. (1) are straightforward: the stream power per unit basal area is the rate at which the potential energy of the water turn into heat; the force resisting the flow is due to the rigid bed; the flow rate per unit channel width is that of the water, q_w; and the cross-sectionally averaged velocity is $q_w/(H_F - \xi)$ so that the friction factor simply reads

$$f_w^* = \frac{8g(H_F - \xi)^3 \sin\phi}{q_w^2} \tag{2}$$

Here, the friction factor is starred to emphasize that it refers to flows in the absence of sediment transport.

In the case of flows over erodible beds in the presence of sediment transport, there are two unambiguous possibilities. From the *water* perspective and in the context of a two-phase approach, the stream power per unit basal area is the rate at which the potential energy of the water is transferred to the sediments; the forces resisting the flow are that at the bed, often negligible given that there the local particle concentration is large and, consequently, turbulence is suppressed (Berzi and Fraccarollo 2015), and the drag on the moving sediments; the flow rate per unit channel width is that of the water q_w; the area occupied by the liquid per unit width is $(H_M - \int_0^{H_M} \nu dy)$; and the cross-sectionally and Favre-averaged velocity is $q_w/(H_M - \int_0^{H_M} \nu dy)$, so that

$$f_w = \frac{8g(H_M - \int_0^{H_M} \nu dy)^3 \sin\phi}{q_w^2} \tag{3}$$

From the *mixture* perspective and in the context of a single-phase approach, the stream power per unit basal area is the rate at which the potential energy of the mixture turns into heat; the force resisting the flow is that at bed; the flow rate per unit channel width is that of the mixture, $q_m = (q_s\rho_s + q_w\rho_w)/\rho_m$, with q_s the volume rate of sediment transport per unit channel width and $\rho_m = [\rho_s \int_0^{H_M} \nu dy + \rho_w \int_0^{H_M} (1-\nu) dy]/H_M$ the mixture density; and the cross-sectionally averaged velocity is q_m/H_M, so that

$$f_m = \frac{8gH_M^3 \sin\phi}{q_m^2} \tag{4}$$

Experimental Set-Up and Conditions

The experiments were carried out at the Water Engineering Laboratory of Czech Technical University in Prague (CTU) in the closed circuit sketched in Fig. 2(a), in which either clear water or a mixture of water and sediments can flow. The total volume flow rate q_m (water plus sediments, if present) was measured through an electromagnetic flowmeter, with an error of ± 0.2 l/s, while pressure transmitters in the vertical pipes of the circuit were employed to estimate the ratio of solid to total flow rate q_s/q_m (Wilson et al. 2006) with an error of 1%. Then, from q_m and q_s/q_m, we determine the water and sediment flow rates, and their associated uncertainties due to error propagation (Table 1). An 8-m long flume of rectangular cross-section (of width W equal to 0.2 m and height equal to 0.25 m), transparent sidewalls, and an adjustable slope is part of the closed circuit.

Profiles of the mean $x-$ component of the velocity (both with and without sediment transport) and solid volume concentration (for the mobile bed case) along the flume midsection were measured 6 meters downstream the flume inlet. The velocity distribution (Bareš et al. 2016) was measured employing an ultrasonic velocity profiler (UVP, Met-Flow SA, 4 MHz). The probe was positioned parallel to the flow and the measures were taken at different elevations; the measuring element volume was a cylinder (5 mm diameter and height of 0.74 mm). Profiles of solid volume concentration, in the presence of sediment transport, were also measured by filming the deformation of a laser stripe penetrating through the flow in the near-wall region (Matoušek et al. 2019). For this purpose, we used a laser sheet orthogonal to the flume and a high-speed camera inclined with respect to the laser. Then, the acquired images were converted into vertical position, distance from the wall, and intensity. For every vertical position, at each frame, we identified the one with the maximum brightness (that corresponds to the particle illuminated by the laser) and then recorded its distance from the wall. The mean free path is the average distance that the laser can travel before hitting a grain: if the flow is dilute, then the averaged distance is high, but if it is concentrated it will be low; that is, through a conversion model (Matoušek et al. 2019) it is possible to estimate local concentration from the mean free path.

The position of the water free surface (± 0.05 cm) was obtained from five ultrasonic gauges along the flume and allowed us to determine the free surface slope and check that was equal to the angle of inclination of the bed (± 0.1 deg), ensuring flow uniformity. A more detailed description of the experimental set up is reported elsewhere (Bareš et al. 2016; Krupička et al. 2018; Matoušek et al. 2015).

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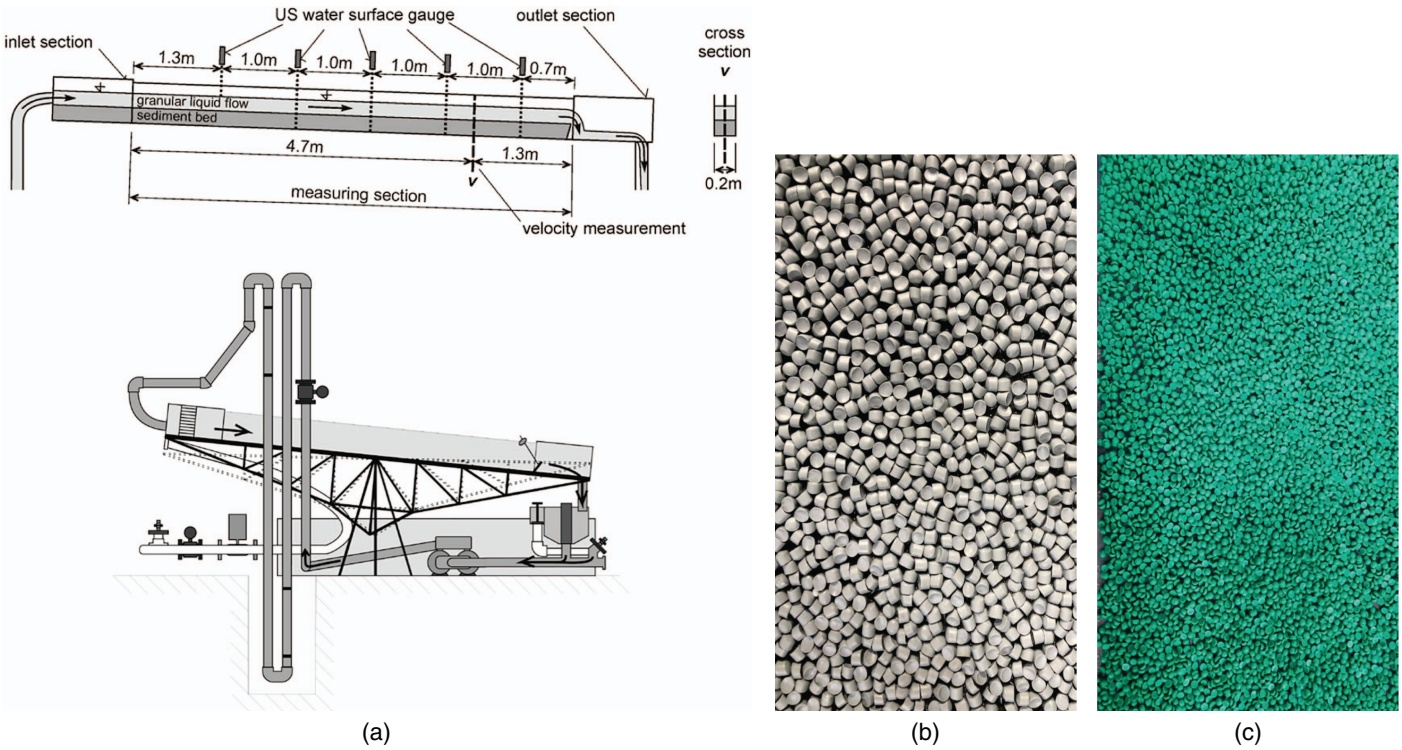

Fig. 2. (Color) (a) Sketch of the recirculating flume; and (b and c) pictures of the rough plates on which either white FA60 particles (b) or green FA30 particles (c) are glued, as uniformly as possible. For FA30 particles, we report the same small protrusion due to the fact that a particle is glued over two other particles.

Table 1. Summary of the experimental results and the associated error

Run	d (mm)	ϕ (degrees)	Q_w (l/s)	Q_s (l/s)	H_M (cm)	$H_F - \xi$ (cm)	ξ (cm)
FA30_1	3.65	1.58 (±0.1)	5.48 (±0.20)	0.36 (±0.01)	3.62 (±0.16)	3.12 (±0.10)	1.13 (±0.21)
FA30_2	3.65	2.03 (±0.1)	5.23 (±0.19)	0.62 (±0.02)	3.79 (±0.14)	2.82 (±0.10)	1.12 (±0.20)
FA30_3	3.65	2.36 (±0.1)	5.26 (±0.18)	0.89 (±0.03)	3.98 (±0.32)	2.74 (±0.10)	1.43 (±0.33)
FA30_4	3.65	2.88 (±0.1)	4.55 (±0.18)	0.93 (±0.04)	3.36 (±0.32)	2.35 (±0.10)	1.34 (±0.43)
FA30_5	3.65	2.09 (±0.1)	5.39 (±0.19)	0.72 (±0.03)	3.87 (±0.17)	2.86 (±0.10)	1.26 (±0.19)
FA30_6	3.65	1.72 (±0.1)	5.76 (±0.20)	0.55 (±0.02)	3.91 (±0.12)	3.12 (±0.10)	0.96 (±0.09)
FA60_1	6.42	1.22 (±0.1)	7.86 (±0.21)	0.26 (±0.01)	5.49 (±0.05)	4.02 (±0.08)	0.86 (±0.03)
FA60_2	6.42	2.54 (±0.1)	7.15 (±0.20)	0.94 (±0.03)	4.73 (±0.14)	3.11 (±0.08)	1.66 (±0.20)
FA60_3	6.42	3.05 (±0.1)	6.76 (±0.19)	1.36 (±0.04)	4.43 (±0.07)	2.88 (±0.08)	1.85 (±0.06)
FA60_4	6.42	3.36 (±0.1)	6.57 (±0.18)	1.73 (±0.05)	4.75 (±0.07)	2.76 (±0.08)	2.35 (±0.13)
FA60_5	6.42	0.68 (±0.1)	10.12 (±0.22)	0.13 (±0.00)	7.16 (±0.05)	5.67 (±0.08)	0.56 (±0.02)

We employed two different types of solid particles during the experimental campaign: plastic lenses, named FA30, of mean diameter 4.0 mm and height 2.2 mm, so that the diameter of the equivalent sphere is $d = 3.65$ mm, density $\rho_s = 1368$ kg/m^3, and volume concentration at rest $\nu_r = 0.60 \pm 0.01$; and plastic cylinders, named FA60, of mean diameter 6.1 mm and height 5.1 mm, so that the diameter of the equivalent sphere is $d = 6.42$ mm, density $\rho_s = 1{,}419$ kg/m^3, and $\nu_r = 0.64 \pm 0.02$. We measured the volume concentration at rest after letting the particles settle in a graduated cylinder full of water, without additional compaction. The experiments were repeated several times to obtain averages and standard deviations.

For flows in the presence of sediment transport, a weir located at the outlet enables to control the thickness of the erodible bed in the flume. For flows in the absence of sediment transport, two series of 2-m plates in length, on which a layer of either FA30 or FA60 particles was glued, were placed at the bottom of the open channel, providing the desired roughness [Figs. 2(b and c)].

We performed a series of 22 experiments on 11 pairs of equivalent configurations, that is, as mentioned, flows over rigid, rough beds in the absence of sediment transport and over erodible beds in the presence of sediment transport, in which the water flow rate and the angle of inclination of the bed were the same. They were chosen so that, in the case of sediment transport: (1) the latter was intense, to emphasize the difference between a single- and a two-phase approach on the evaluation of the friction factor, but the sediments were not suspended by turbulence; and (2) the analysis was not confounded by the presence of bedforms or other kind of surface instabilities. This regime of sediment transport has been termed in the literature as the upper plane bed regime (Matoušek et al. 2013), sheet flow (Jenkins and Hanes 1998), or collisional suspension (Berzi and Fraccarollo 2013). To exclude the possibility of turbulent suspension, we have checked that the ratio of the single particle settling velocity to the water shear velocity at the top of the particles, that is the square root of the water shear stress per unit fluid density there, was larger than unity (Berzi and Fraccarollo 2016).

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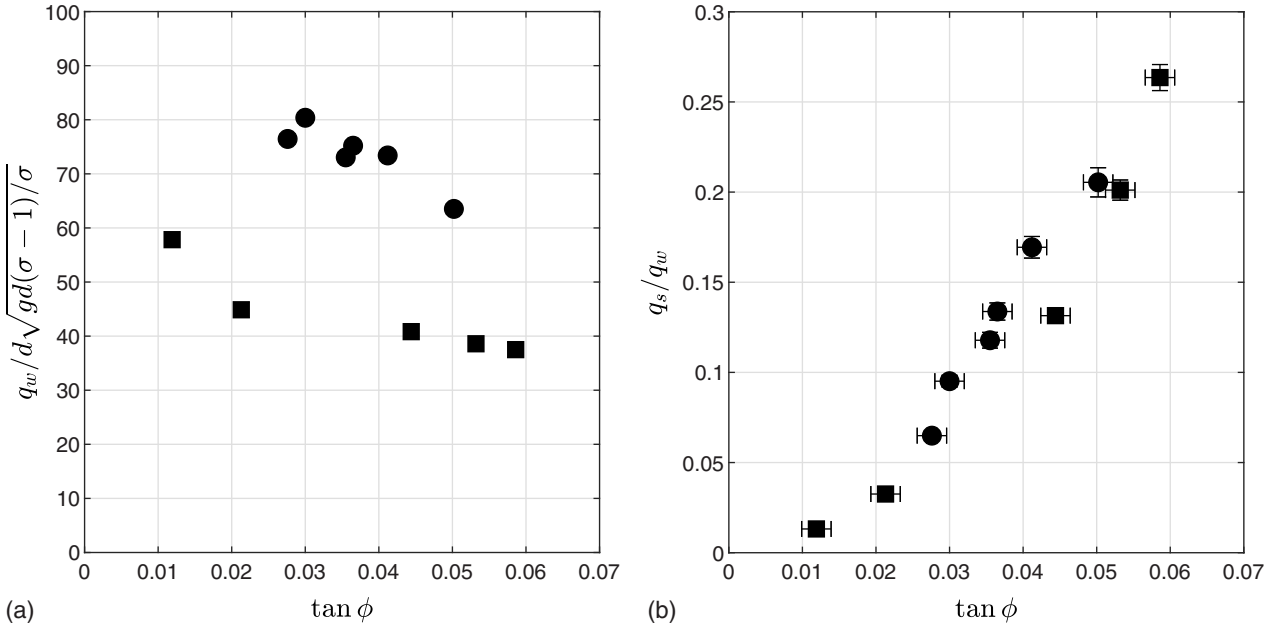

Fig. 3. (a) Dimensionless water flow rate as a function of the bed slope; and (b) ratio of sediment over water flow rate for flows over erodible beds in the presence of sediment transport as a function of the bed slope. Particles FA30 (circles) and particles FA60 (squares).

Fig. 3(a) shows the investigated values of water flow rate per unit channel width and bed slope for the two types of sediments. In what follows, quantities are made dimensionless using the particle equivalent diameter, d, and the reduced gravitational acceleration, $g(\sigma-1)/\sigma$, where σ is the ratio of particle-to-water mass densities.

Fig. 3(b) depicts the measured ratio q_s/q_w as a function of the bed slope in the case of flows over erodible beds. In the case of intense sediment transport, since the particle density is similar, the data collapses onto a single curve (Berzi and Fraccarollo 2013). This confirms that the water flow rate and the bed slope are suitable control parameters and that the sediment flow rate adjusts automatically to reach a steady and fully developed configuration. We emphasize that the sediment transport rate in our experiments on erodible beds is much higher than that investigated in most of the literature (see Recking et al. 2008b) on the effect of sediment transport on flow resistance.

Fig. 4 shows examples of measured profiles of sediment concentration and mean velocity in the flow direction for one of the flows over erodible beds involving FA30 particles. As mentioned, ultrasonic gauges permit to identify the position of the free surface. The position at which the linear interpolation between the two closest values of the moving average (over a window of three data points) of the concentration profile reaches the value ν_r permits the determination of the top of the erodible bed, according to the definition of (Berzi et al. 2019). Other authors (Taberlet et al. 2003) have suggested to take the top of the erodible bed at the intercept between the y-axis and the linear part of the profile of the mean velocity in the flow direction [Fig. 4(b)]. The different positions of the erodible bed evaluated according to these two definitions are reported in Fig. 4. Their average corresponds to the location at which we set $y=0$ in our experiments, while half of their difference provides an estimate of the associated uncertainty. Given that the concentration is evaluated in the near-wall region, while the velocity is measured along the channel mid-section, this procedure permits to account for, at least partially, the variability of the geometry of the erodible bed in the spanwise direction (Jop et al. 2006).

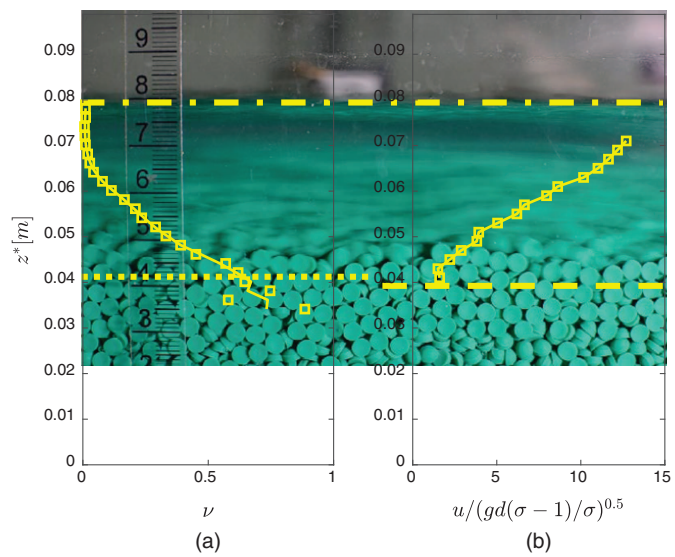

Fig. 4. (Color) Measured profiles (symbols) of (a) concentration; and (b) mean velocity in the flow direction for one of the experiments on flows over erodible beds with FA30 particles, superimposed on one frame displaying the near-wall region of the flow. Here, the origin of the vertical coordinate z^* is taken at the position of the rigid base below the erodible bed. Also shown are the moving-average of the concentration profile (solid line), the location of the free surface (dot-dashed line), and the top of the bed according the concentration (dotted line) and velocity method (dashes line).

For flows over erodible beds, the uncertainty in the flow depth H_M depends on the errors associated with the positions of both the water surface and the erodible bed. The uncertainty in the determination of $\xi=\int_0^{H_M}\nu dy/\nu_r$ depends on the errors associated with the position of the erodible bed and the volume concentration at rest (given that there are no particles near the water surface, its

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uncertainty has no consequence on the evaluation of the integral of the volume concentration in the definition of ξ).

For flows over rigid, rough beds, the uncertainty in the flow depth $H_F - \xi$ is due to the error associated with the position of the water surface and to the one associated with the position of the bed. In this case, the bed is positioned at the top of the equivalent diameter, and its uncertainty is computed as half of the difference between the two principal diameters of the particle.

The characteristics of the experiments and the measurements are summarized in Table 1.

Results and Discussion

The values of the friction factors that we have defined in the previous section are reported in Table 2 and in Fig. 5 as functions of the bed slope. We first notice that the average values of f_m are always larger than those of f_w and that, in all but one case, the average values of f_w are larger than those of f_w^*. This indicates that the flow resistance experienced by the mixture of water and sediment is higher than the flow resistance experienced solely by the water (mainly due to the drag acting on the particles) in the case of flows over erodible beds with sediment transport. It also suggests that, in general, the presence of sediment transport increases the flow resistance from the water perspective ($f_w > f_w^*$). However, our conclusions are weakened in the case of FA30 particles by the superposition of some error bars in Fig. 5(a), mainly due to the uncertainty in the identification of the location of the erodible bed. This uncertainty might also explain the apparently contradictory statements present in the literature about the role of sediment transport on flow resistance.

Table 2. Shields parameter and friction factors, with the associated error

Run	θ	f_{w*}	f_w	f_m
FA30_1	0.98 (±0.2)	0.09 (±0.01)	0.07 (±0.02)	0.13 (±0.02)
FA30_2	1.29 (±0.2)	0.09 (±0.01)	0.12 (±0.02)	0.19 (±0.03)
FA30_3	1.68 (±0.4)	0.09 (±0.01)	0.14 (±0.05)	0.23 (±0.06)
FA30_4	1.80 (±0.6)	0.10 (±0.01)	0.13 (±0.06)	0.21 (±0.06)
FA30_5	1.40 (±0.2)	0.09 (±0.01)	0.12 (±0.03)	0.19 (±0.03)
FA30_6	1.08 (±0.1)	0.09 (±0.01)	0.10 (±0.02)	0.15 (±0.02)
FA60_1	0.50 (±0.0)	0.07 (±0.01)	0.13 (±0.01)	0.18 (±0.02)
FA60_2	1.10 (±0.1)	0.08 (±0.01)	0.13 (±0.02)	0.24 (±0.03)
FA60_3	1.33 (±0.0)	0.09 (±0.01)	0.13 (±0.01)	0.24 (±0.02)
FA60_4	1.71 (±0.1)	0.09 (±0.01)	0.15 (±0.02)	0.31 (±0.02)
FA60_5	0.34 (±0.0)	0.07 (±0.01)	0.11 (±0.02)	0.13 (±0.02)

The friction factor f_w^* relative to the water, in the case of flows over rigid, rough beds without sediment transport, is weakly dependent on the bed slope, for roughness provided by either FA30 or FA60 particles. The friction factor f_w relative to the water, in the case of flows over erodible beds with sediment transport, increases with bed slope and, then, seemingly reaches a plateau for FA30 particles [Fig. 5(a)]. Instead, it is roughly independent of the bed slope for FA60 particles [Fig. 5(b)]. The friction factor f_m relative to the mixture, in the case of flows over erodible beds with sediment transport, almost always increases with bed slope for both types of solid particles, as in (Wilson and Nnadi 1992).

The three friction factors that we have identified approach each other at the lowest values of the bed slope, and this is more evident for the fine particles. Lower values of bed slope correspond to lower values of the ratio of the sediment and water flow rates [Fig. 3(b)]. Hence, intuitively, if the sediment transport is weak, say the sediment flow rate is less than 5% than that of the water, it becomes a difficult task to distinguish between single- and two-phase approaches or between flows over rigid, rough, and erodible beds.

In the case of flows over rough, rigid beds in the absence of sediment transport, we can imagine that the turbulence is governed by a characteristic roughness, r, of the order of the particle diameter d. Indeed, the associated boundary Reynolds number, $\rho_w d\sqrt{g(H_F - \xi)\sin\phi}/\mu$, is much larger than 100 in all our experiments, so that the flow is hydraulically rough. Then, the friction factor must only be a function of the relative roughness r/R, where R is the hydraulic radius. There are a number of empirical relationships between the friction factor and the relative roughness in the

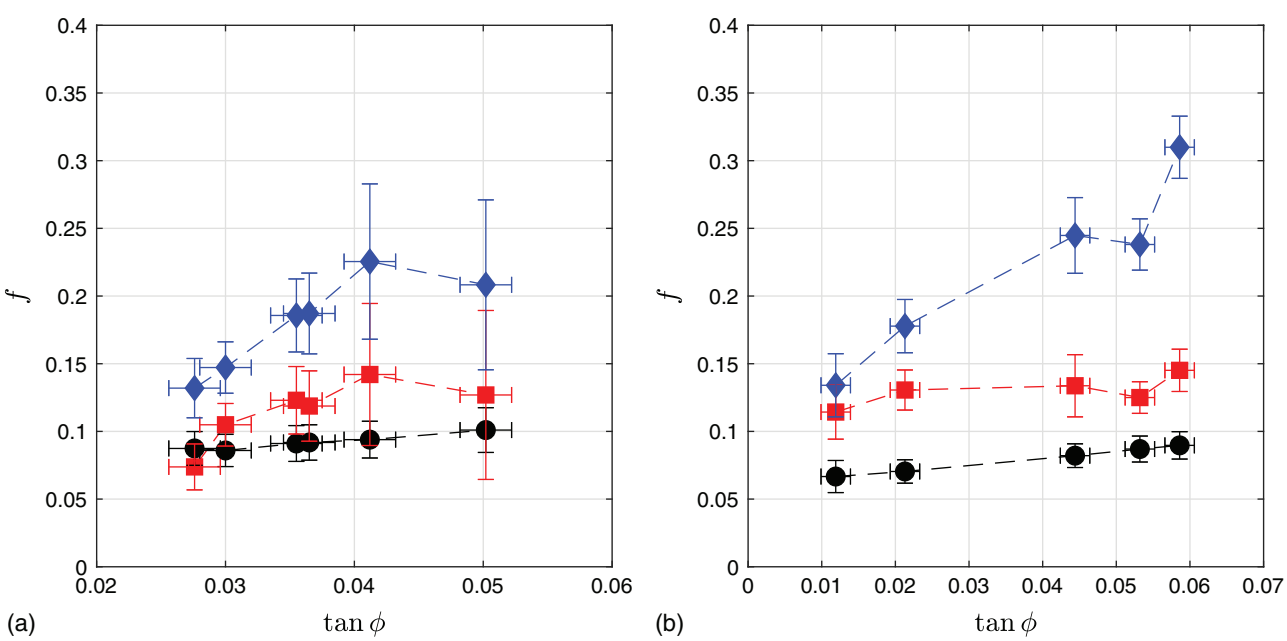

Fig. 5. (Color) Friction factors f_w^* (black circles), f_w (red squares), and f_m (blue diamonds) as functions of the bed slope for: (a) particles FA30; and (b) particles FA60. Note that the range of the horizontal axis is not the same.

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literature (Ferguson 2007). Recking et al. (2008a) report that the logarithmic function is a good approximation both for fixed and mobile bed flow (up to a Shields parameter equal to 0.25)

$$\left(\frac{8}{f}\right)^{1/2} = 6 + \frac{1}{\kappa}\ln\frac{R}{r} \tag{5}$$

where κ is the von Kármán's constant.

Many scholars (Abrahams and Li 1998; Banerjee et al. 2019; Hu and Abrahams 2004; Simoes 2010) have successfully applied similar friction laws also in the presence of sediment transport. However, we emphasize that this choice is unphysical, given that the mechanisms at the origin of the flow resistance when sediment transport is present are different than the generation of turbulent eddies near the rough base that grow in size as they move further from the boundary. At least in the case of intense sediment transport without turbulent suspension, the turbulent eddies originate at the particle surface and are confined in the space available between the solid particles, so that their size is only a fraction of the sediment diameter. Most of the flow resistance from the water perspective is due to the drag exerted on the particle surfaces and the momentum exchange between the fluid shells surrounding the particles themselves, while the flow resistance from the particle perspective is due to momentum exchange in collisions (Berzi and Fraccarollo 2015). However, from practical engineering purposes, it is worthwhile to check whether Eq. (5) is able to reproduce the behavior of flows with intense sediment transport, with appropriate choices for the roughness r.

Recking et al. (2008a) performed 134 experiments comparing flows with and without sediment transport. They showed that Eq. (5), with $\kappa = 0.31$ and the hydraulic radius corrected for the presence of the side wall (Vanoni and Brooks 1957), is able to reproduce f_w^*, if $r = 2.1d$, and f_w, if $r = 5.3d$.

In our definitions of the friction factor, we neglect the role of sidewalls in determining the hydraulic radius (Guo 2017), given that, for flows without sediment transport, the corrected friction factor would be within the error bars of Fig. 5. Also, the sidewall corrections would not be applicable to the case of intense sediment transport, where the presence of moving particles would alter the velocity profile and modulate the turbulence in proximity to the lateral walls, and make the comparison with flows without sediment transport less straightforward.

The friction law suggested by (Recking et al. 2008a) can describe our clear water experiments, if the hydraulic radius is simply calculated as $R = H_F - \xi$, $\kappa = 0.41$, and r equal to 1 and 2 particle diameters, respectively, when the roughness is provided by either FA60 or FA30 plastic particles [Fig. 6(a)]. The larger roughness coefficient associated with the smaller particles (FA30) is likely due to the fact that, for such small non spherical grains, the gluing process at the base is more difficult, and localized patches of height larger than one particle diameter are inevitable.

If we plot $(8/f_m)^{1/2}$ in the presence of sediment transport as function of the relative submergence R/d, where $R = H_M$, the measurements follow reasonably well Eq. (5), with $\kappa = 0.41$ and $r = 7$ particle diameters, for both FA30 and FA60 particles [Fig. 6(a)].

Interestingly, the friction factor f_w from the water perspective in the presence of sediment transport seems to be roughly constant, independent of the relative submergence and, therefore, of the amount of conveyed solid material [Fig. 6(a)].

Fig. 6(b) depicts the comparisons between Eq. (5) and the measurements of (Recking et al. 2008a) (where $\sigma = 2.6$ and $d = 2.3, 4.9, 9$ and 12.5 mm). For consistency, we have not corrected the friction factor to account for the presence of sidewalls. For inclined flows over erodible substrates in the absence of sediment transport, we have removed all data with Shields number larger

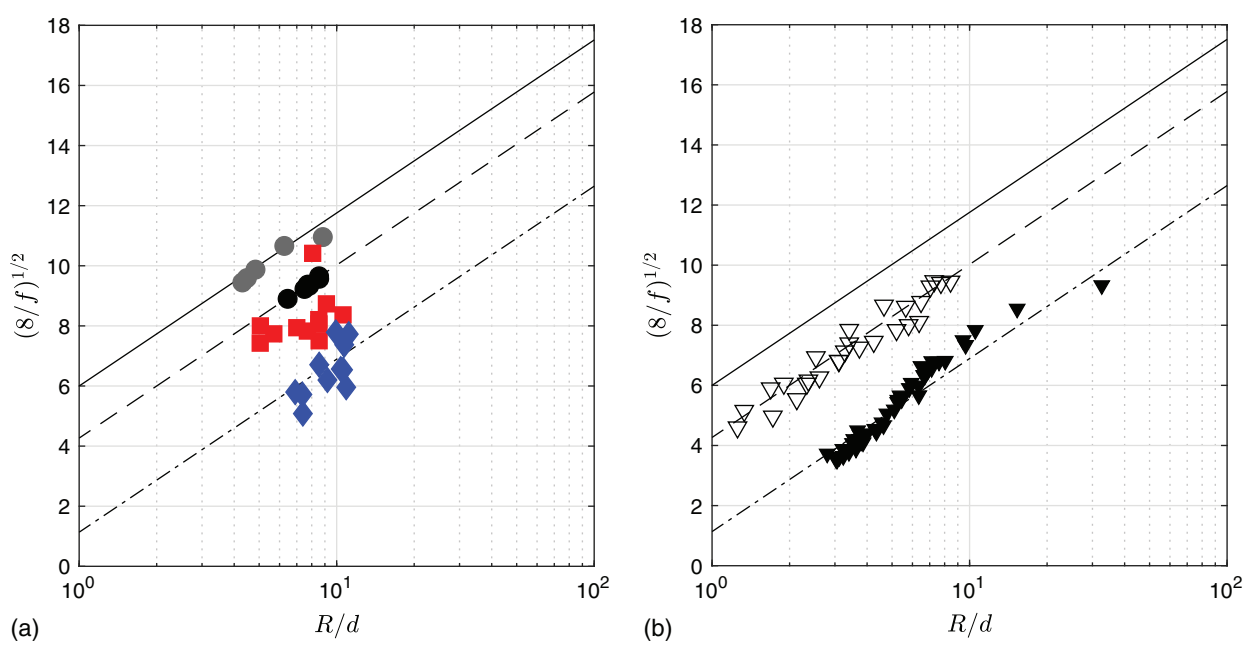

Fig. 6. (Color) (a) present experiment; and (b) experiment after (Recking et al. 2008a); measurements of $(8/f)^{1/2}$ as functions of relative submergence, when the friction factor is: f_w^* in the absence of sediment transport over rigid, rough beds with roughness provided by FA30 particles (black circles) and FA60 particles (grey circles); f_w (red squares) and f_m (blue diamonds) in the presence of sediment transport, for both types of plastic sediments; f_w^* (void triangles) in the absence and $f_w \approx f_m$ (filled triangles) in the presence of transport of well-sorted gravel over erodible beds consisting of the same material. Lines represent Eq. (5) when the roughness coefficient is taken to be 1 (solid line), 2 (dashed line), and 7 (dot-dashed line) particle diameters.

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than 0.045, to eliminate the possibility of any undetected, weak sediment flow. For inclined flows over erodible beds with transport of gravel, we have considered all data with Shields number greater than 0.15 to include only measurements of intense sediment transport (Berzi and Fraccarollo 2013) and enable a direct comparison with our data. The ratio q_s/q_w in (Recking et al. 2008a) is less than 4%, whereas in our experiments it is in the range between 1% and 20%. At small values of q_s/q_w, $f_w \approx f_m$. Fig. 6(b), then, shows that f_w actually depends on the relative submergence at low flow rate ratios.

Figs. 6(a and b) indicate that Eq. (5) is able to reproduce the behavior of flows with and without sediment transport for a broad range of particle density ($\sigma = 1.37$ to $\sigma = 2.60$) and particle size ($d = 2.3$ to 12.5 mm).

As explained above, in dimensionless terms, the thermodynamic and mechanical points of view on flow resistance coincide and are embedded in the friction factor f. However, we must emphasize that, in dimensional terms, the same flow resistance can be associated with different friction factors, for different choices of the normalization variables. Also, depending on which variables are kept constant when comparing two equivalent flows, the mechanical resistance can remain unchanged even though the thermodynamic one varies, and vice-versa. For example, if we look at the water phase only, our equivalent configurations are characterized by the same energy dissipation; however, the resisting force that balances the water weight in the flow direction is larger in the mobile-bed condition, given that, for the same water flow rate, the water velocity is lower and its volume larger.

Finally, we plot the measured values of H_F and H_M in Fig. 7. While we can now reasonably claim that the presence of sediment transport increases the dimensionless flow resistance both from the water and the mixture perspective, it seems, on the other hand, that allowing for the particle motion has very little effect on the water surface elevation with respect to a common reference level. This may be due to the opposite effects of sediment mobilization: on one hand, it increases the friction factor, thus imposing a larger water depth to maintain the same energy slope; on the other, it lowers the base of the flow. Hence, larger flow depths do not necessarily imply higher water surface elevations. These results can be considered valid within the experimental range considered, i.e. for intense sediment transport without turbulent suspension, with mixture Shields parameters ranging between 0.34 and 1.80. For similar but less intense flows (say, Shields between 0.15 and 0.25) (Recking et al. 2008a) reports a clear increase of f_w: in this case, being the transport layer small, it might be that the offset ξ is not big enough to compensate for the increase of flow depth, leading to a non-negligible modification of the water surface elevation.

The main results of this work can be summarized as follows: the presence of intense sediment transport over a plane bed (without bed forms), with respect to an equivalent clear water flow, increases the friction factor and the flow depth, but might leave the water surface elevation unaltered due to the lowering of bed position.

We now examine the consequence of this phenomenological finding on river engineering modelling (see Fig. 8). Let us take the simple case of defining a discharge-stage curve $z(Q_w)$ for a river reach of slope $S_0 = 0.01$ under spatially uniform conditions. For low discharges we do not have sediment transport, so that $z(Q_w) = z_b + h(Q_w)$, where z_b is the elevation of the (immobile) top of the bed and h is the water depth. The latter would be computed by a typical normal depth relation (under the approximation for wide rectangular channels): $h \approx (nQ_w/W)^{3/5} S_0^{-3/10}$, where $n = 0.033$ [s/m$^{1/3}$] is the Manning coefficient.

For higher values of Q_w the river reach will be characterized by sediment transport, which we will consider to be also uniform in space, so that the equilibrium bed slope remains unchanged, although under dynamic conditions. While computing the corresponding part of the discharge-stage curve, we should now account for the effects of sediment transport and, in particular, of the consequent increase in flow resistance (see introduction). Although present results do not provide indications for intermediate values of the discharge, we have confirmed that for high discharges, corresponding to intense sediment transport, the friction factor f_w increases significantly with respect to the corresponding

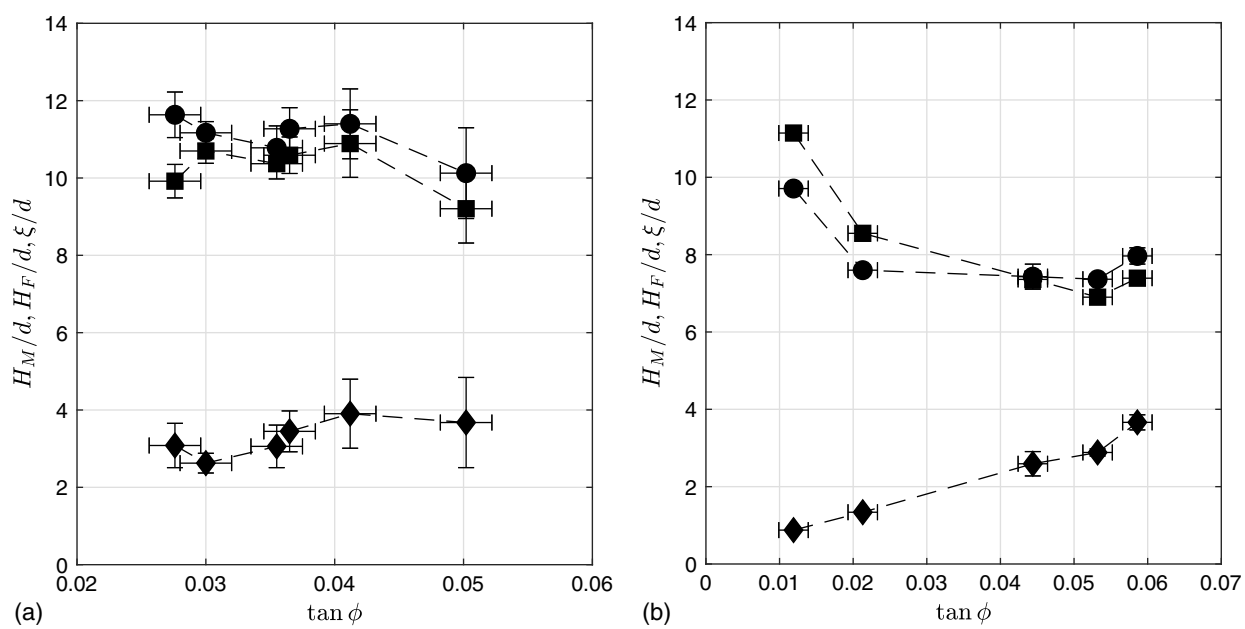

Fig. 7. Free surface elevations as functions of the bed slope in case of flows over rigid, rough beds in the absence of sediment transport (circles) and over erodible beds in the presence of sediment transport (squares), and thickness ξ (diamonds) for: (a) FA30 particles; and (b) FA60 particles. Note that the range of the horizontal axis is not the same.

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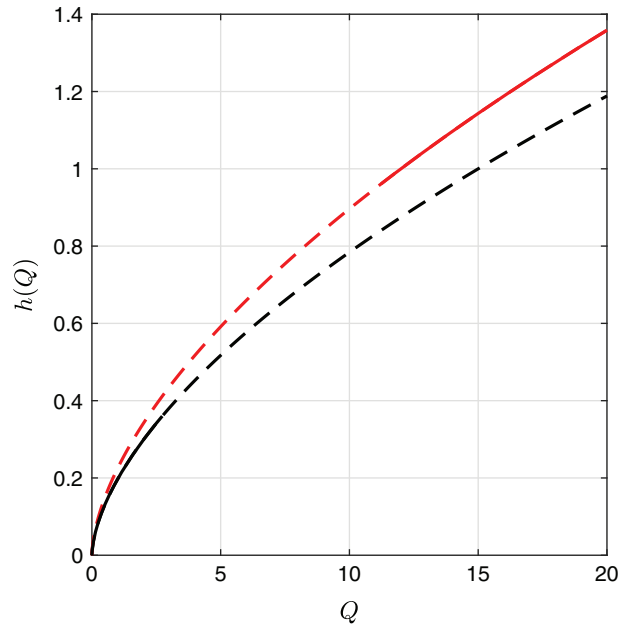

Fig. 8. (Color) Example of stage-discharge curve for a mountain torrent [slope 1% and $W = 5$ (m)]. Black curve: $n = 0.033$ (s/m$^{1/3}$), clear water flow; Red curve: $n_1 = 0.042$ [s/m$^{1/3}$], mobile bed flow with intense transport of sediment. The lines are continuous if evaluated in their range of application: Shields numbers less than 0.045 (black) and greater than 0.15 (red).

conditions without transport, f_w^*: our results suggest $f_w/f_w^* = 1.40$, comparable to data from (Gao and Abrahams 2004) ($f_w/f_w^* = 1.45$) and to the model of (Recking et al. 2008b) ($f_w/f_w^* = 1.60$). By taking a reference value $f_w/f_w^* = 1.50$ we obtain an increase of the Manning coefficient of 1.25 (leading to $n_1 = 0.042$ [s/m$^{1/3}$]) and a consequent increasing factor of 1.15 for the water depth with respect to the trend of the stage curve for immobile bed. However, our results suggest that the water elevation is not correspondingly changing, as the increase of h is compensated by a decrease of z_b. If the selected hydraulic model is a typical fixed bed one, then the modeler should increase the roughness coefficient for the depth computation (Fig. 8), but leave it unchanged for calculating the elevation.

Another option is given by the mixture perspective. In this case, a proper estimate of the flow resistance is f_m, which allows us to compute H_M. Under this scenario, however, it is necessary to account for the lowering of the bed position, otherwise the water surface elevation might be overestimated. However, due to the limited number of experiments, we are not able to provide closure for ξ in this work. Eventually, the modeller should adopt a two-phase flow approach. In this case, more refined models are available, such as the one based on kinetic theory of granular gases (e.g., Berzi and Fraccarollo 2013; Matoušek and Zrostlík 2020): they have an advantage to be physically based and provide detailed information about the flow structure.

Conclusions

In this work, we studied the effect of sediment transport on flow resistance and water surface elevation in an open channel with a granular bed. We addressed this problem by arranging an experimental comparison of two mutually related flows of the same water flowrate and the bed slope—one with and the other without transport of bed sediment. Previous literature studies agree on the intuitive result that the bed shear stress increases as a result of the transport of sediment interacting with the bed. However, the same is not necessarily true for corresponding bed friction factors, due also to possible differences in their definitions.

We proposed definitions for three different friction factors: one for the water in the absence of sediment transport, one for the water in the presence of sediment transport, and one for the mixture; such quantities are consistent from both a mechanical and a thermodynamic point of view.

Following the proposed definitions, we obtained that the friction factors for the water and for the mixture in sediment-laden flow are higher than the friction factor for the equivalent clear water flow. This result is consistent with most of the cited literature; at the same time, ambiguities in the definitions can explain different results proposed in other works.

We found that the presence of sediment transport increases the flow depth and simultaneously, sediment mobilization leads to a decrease in the bed surface position, thus providing more space through which the water can flow. Consequently, the water surface elevation seems to remain unaffected by the presence of sediment transport. Finally, we proposed how engineers can account for this finding in practical computations of, e.g., discharge-stage curves in channels with mobile beds.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository or online in accordance with funder data retention policies. (https://www.researchgate.net/publication/359495805_data_Experimental_comparison_of_inclined_flows_with_and_without_intense_sediment_transport_flow_resistance_and_surface_elevation).

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Notation

The following symbols are used in this paper:

d = particle mean diameter (m);
f = friction factor;
g = gravitational acceleration (m/s^2);
H_F = water surface elevation for fixed bed flows over rigid, rough beds (m);
H_M = water surface elevation for flows over erodible beds (m);
q_w = water flow rate per unit channel width (m^2/s);
q_s = solid flow rate per unit channel width (m^2/s);
R = hydraulic radius (m);
r = roughness coefficient (m);
u = local mean velocity in the flow direction (m/s);
W = channel width (m);
ξ = depth of the sediment layer at rest (m);
μ = water viscosity ([kg/ms]);

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ν = local volume concentration of sediments;
ν_r = sediment volume concentration at rest;
ρ_w = particle density (kg/m^3);
ρ_s = water density (kg/m^3); and
σ = particle-to-water density ratio.

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